

Hardware, software and AI for STEM education — engineered end-to-end, under one roof.

Curio Makerlabs builds a complete learning ecosystem: an industrial-grade controller, a modular PCB construction system, a plug-and-play sensor library, an in-house coding studio, and AI tools with which students train and deploy real machine-learning models. One platform carries a learner from a first circuit to a working, real-world product.

01 OUR OWN HARDWARE

CPU, sensors and structural parts designed and manufactured by Curio — not re-badged imported kits.

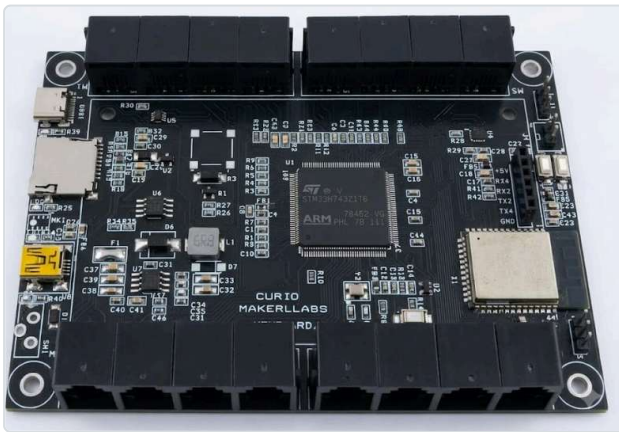
02 OUR OWN SOFTWARE

A complete coding studio developed in-house: blocks to text code, live data, wireless programming, teacher tools.

03 OUR OWN AI

Model training for image, voice, pose and gesture; an LLM copilot; and a RAG-based AI mentor for every student.

THE CURIO CPU



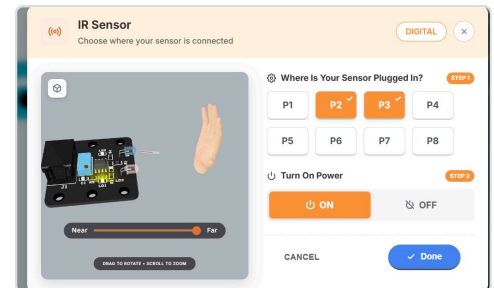
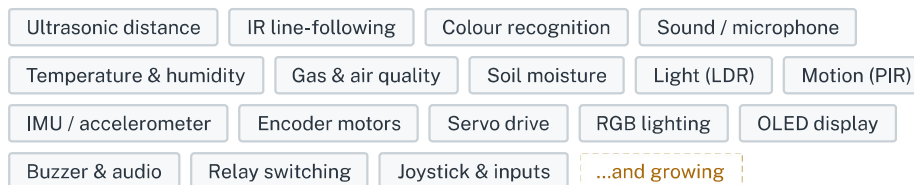
CURIO CPU — DESIGNED, ENGINEERED & ASSEMBLED IN-HOUSE

One controller, engineered for classrooms

PROCESSOR	Industrial-grade 32-bit ARM Cortex-M7 @ 480 MHz — real-time headroom for robotics, control loops and on-device AI workloads.
CONNECTIVITY	Onboard Wi-Fi + Bluetooth for wireless programming, live dashboards, app control and IoT projects.
PLUG & PLAY I/O	16 RJ sensor ports with polarised connectors — no jumper wires, no wiring errors, no short circuits in class.
EXPANSION	USB-C programming, microSD data logging, UART breakout and headers for advanced builds.
BUILT TO LAST	Protected onboard power, classroom-safe low voltage, rugged connectors rated for daily student use.

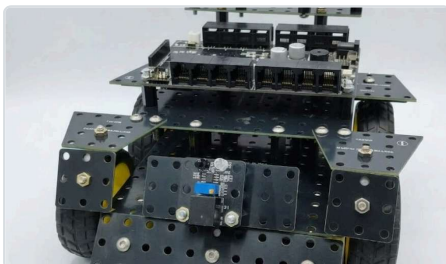
PLUG-AND-PLAY SENSOR ECOSYSTEM

Every module clicks into the same RJ ports — no soldering, no loose wires, no damaged pins. In our software, students pick the port on an interactive 3D view of each sensor, so wiring becomes a ten-second step.



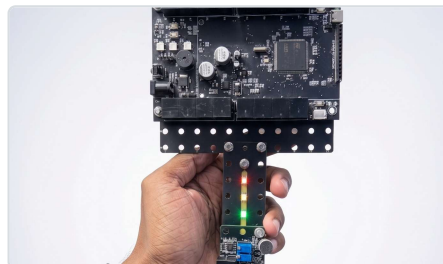
IN-SOFTWARE SETUP — PICK A PORT ON A LIVE 3D SENSOR

FROM KIT TO REAL-WORLD PRODUCT



Gesture-controlled smart car

Avoids obstacles with ultrasonic sensing and steers by hand gestures through a student-trained AI model — a complete real-time control loop.



Smart blind stick

An assistive-technology build that detects obstacles and warns with sound and light — students engineering for the visually impaired.

The chassis is a circuit board.

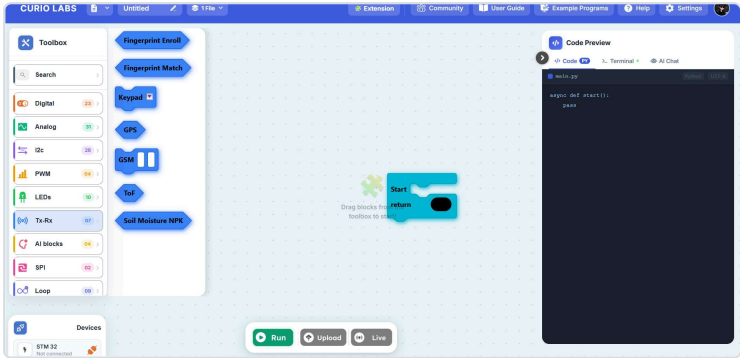
Plates, beams and brackets are precision PCB parts — students bolt together robots the way engineers prototype real products. Every build disassembles back into the kit, ready for the next idea.

One kit, unlimited possibilities

- Agriculture — soil, irrigation & weather monitors
- Environment — air-quality & pollution stations
- Smart city — traffic, lighting & parking models
- Health & accessibility — assistive devices
- Automation — voice & app-controlled systems

OUR SOFTWARE – BUILT ENTIRELY IN-HOUSE

Every screen of our platform is written by our own engineers, so it evolves with our hardware, our curriculum and feedback from real classrooms — no third-party licences, no lock-in.



BLOCK PROGRAMMING WITH LIVE CODE PREVIEW, TERMINAL & AI CHAT

Blocks → real code. Drag-and-drop coding with a live code preview — students graduate to text programming in the same window.

Run · Upload · Live. One-click flashing over Wi-Fi or USB and a live mode for instant feedback — no driver setup on school computers.

A deep toolbox. Digital, analog, I²C, PWM, displays, GPS, GSM, fingerprint, soil NPK, AI blocks and more — every module has its block.

Terminal & live data. A real terminal and streaming sensor readings — students see the data behind every project.

Examples, guides & community. Ready programs, a full user guide and a student community, one click away.

Classroom-friendly. Lightweight and low-bandwidth — built for ordinary school computers, not high-end labs.

THE AI LAB – STUDENTS TRAIN REAL MODELS

A Teachable-Machine-style workflow built into our software: collect samples, train a model in minutes, and deploy it straight to the build. Students don't just use AI — they make it, and watch it drive motors, lights and robots in real time.

Image

Camera classification — waste sorting, quality checks, smart door systems.

Voice

Speech commands & sound recognition — voice-controlled robots and alerts.

Pose

Full-body pose detection — posture alerts, fitness trainers, interactive games.

Gesture

Hand-gesture models that steer cars, arms and lighting — touch-free control.

LLM INTEGRATION

A coding copilot inside the studio

Integrate Large Language Models (LLMs) into your software architecture to transform static telemetry into an intelligent, adaptive decision system. By passing structured sensor variables (e.g., temperature, acceleration, pressure, ambient noise) directly into an LLM context window alongside operational constraints, the engine moves beyond traditional rule-based thresholds to deliver real-time operational intelligence.

RAG-BASED AI MENTOR

Curio Mentor — a private tutor for every student

A retrieval-augmented AI grounded in our own curriculum, project guides and kit documentation — so answers are accurate, age-appropriate and syllabus-aligned. It guides with hints and next steps rather than ready-made answers, is available round the clock, and lets one instructor support an entire lab.

WHAT A CURIO DEPLOYMENT INCLUDES

Kits & spares

Classroom sets with spare-part support

Curriculum

Graded coursework mapped to class levels

Teacher training

Onboarding & certification for instructors

Software & AI

Coding studio, AI Lab & Mentor for every seat

Support

On-ground and remote assistance

Outcomes

Progress reporting for administrators

WHY CURIO FOR PUBLIC EDUCATION

NEP 2020 ALIGNED

Experiential, hands-on learning — coding, AI and 21st-century skills from middle school onward.

ATL & LAB READY

A drop-in platform for Atal Tinkering Labs, robotics labs and skill centres — one vendor, the complete stack.

MAKE IN INDIA

Hardware designed and manufactured indigenously; software and AI developed in-house — self-reliant supply, fast repair.

BUILT TO SCALE

Standardised kits, teacher certification and remotely manageable software make district-wide rollouts practical.

We would welcome the opportunity to demonstrate the platform in person — pilot deployments can be running in selected schools within weeks.